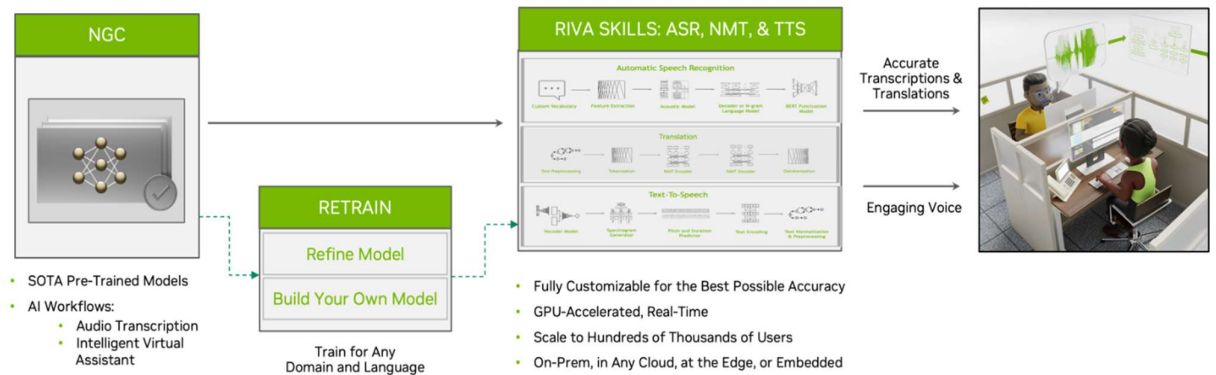


NVIDIA NEMO SPEECH AI RELEVANT STREAMING ARCHITECTURE BACKBONES

1. NVIDIA RIVA REAL TIME MODULAR SPEECH PIPELINE ARCHITECTURE

NVIDIA RIVA

Fully Customizable, GPU-Accelerated SDK for Real-Time Speech and Translation AI

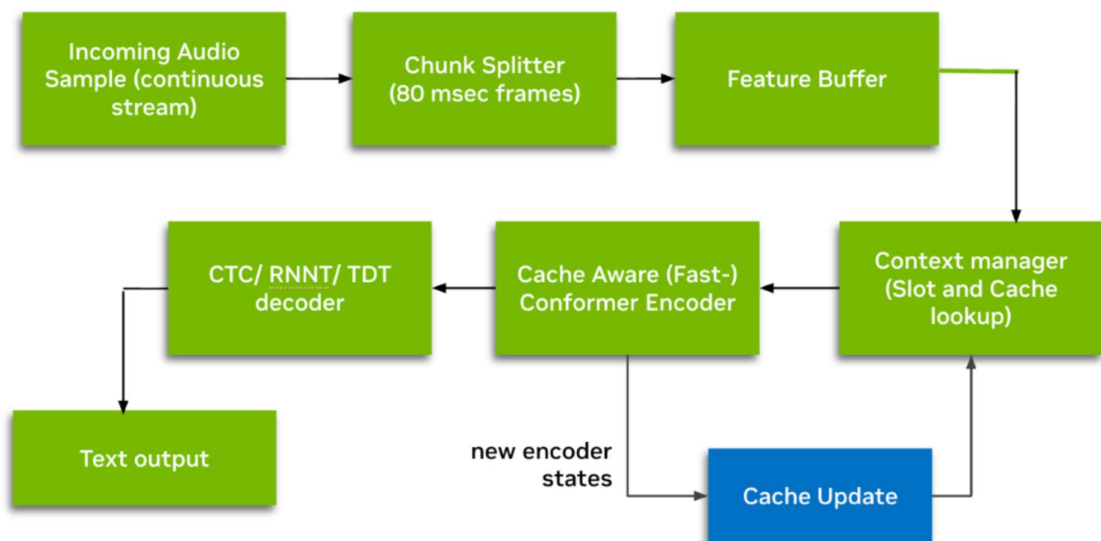


SOTA – State-Of-The-Art | ASR – Automatic Speech Recognition | NMT – Neural Machine Translation | TTS – Text-To-Speech

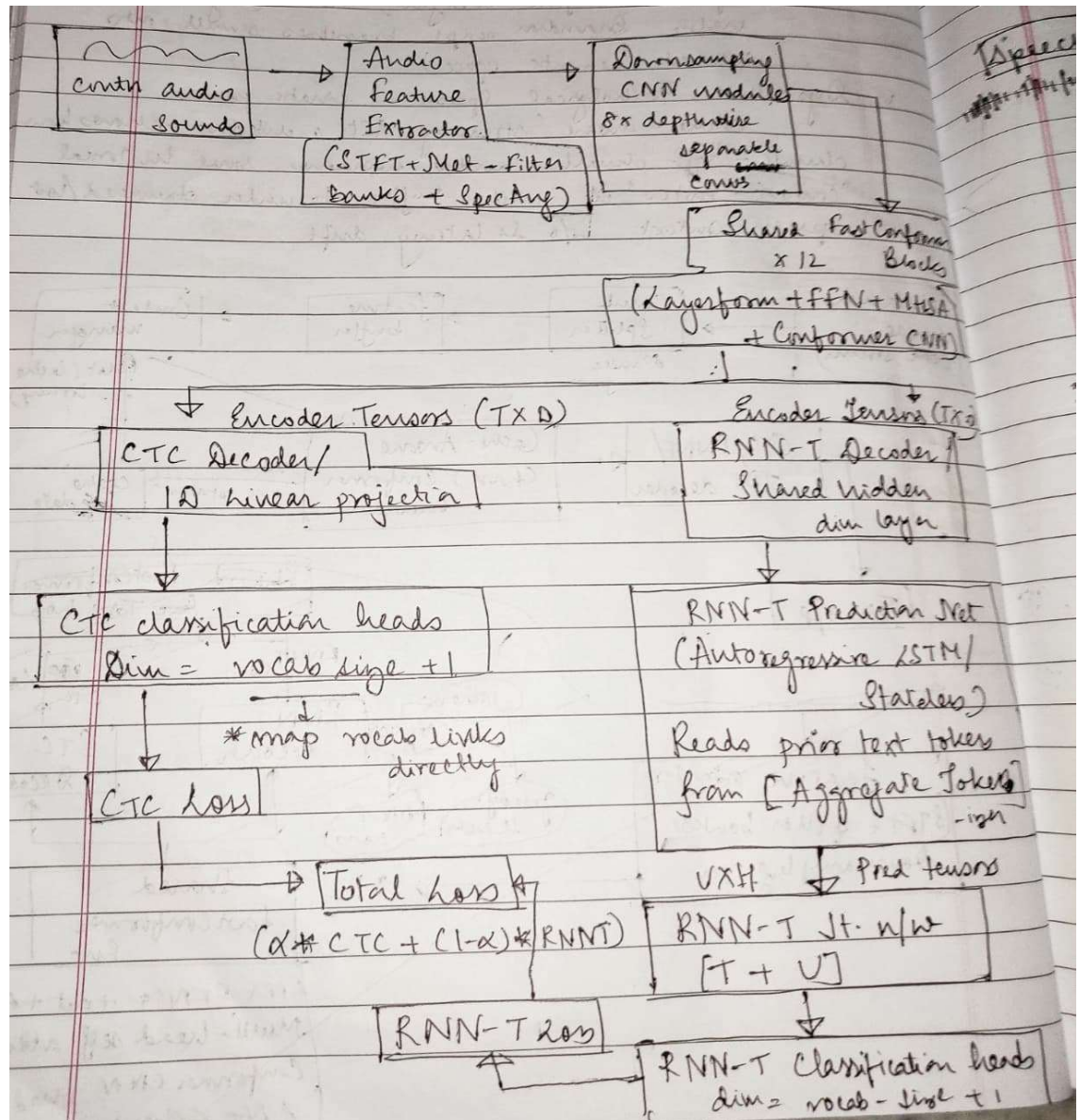
Source: <https://docs.nvidia.com/deeplearning/riva/user-guide/docs/overview.html>

1. FAST CONFORMER-TRANSDUCER AND CACHE AWARE STREAMING

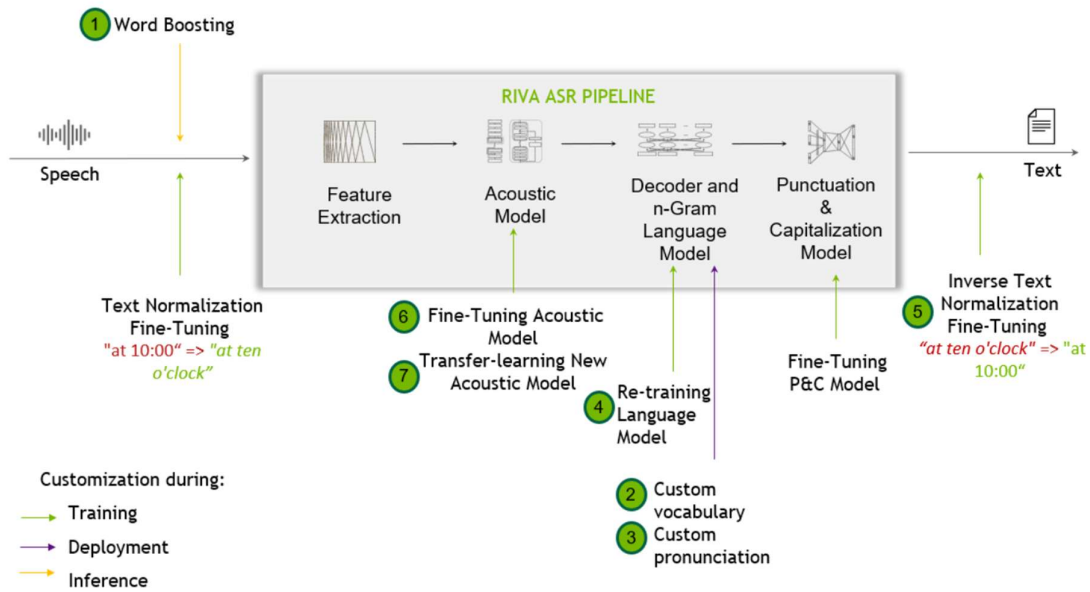
Cache-Aware Streaming Pipeline



Source: <https://huggingface.co/blog/nvidia/nemotron-speech-asr-scaling-voice-agents>

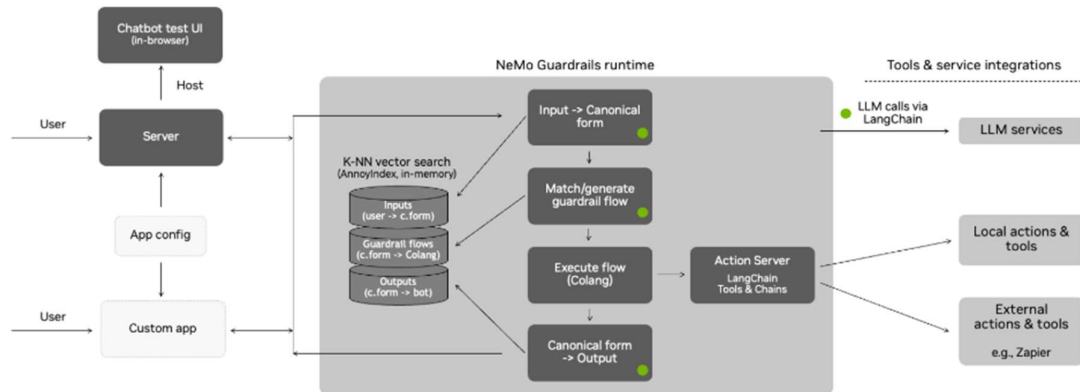


2. RIVA ASR CUSTOMIZATION FOR CUSTOM VOCAB AND LM ADAPTION



Source: <https://docs.nvidia.com/deeplearning/riva/archives/2-10-0/user-guide/docs/asr/asr-customizing.html>

3. NEMO GUARDRAILS AND S2S AGENT ARCHITECTURE



Source: <https://docs.nvidia.com/nemo/guardrails/0.19.0/architecture/README.html>

OpenAI Whisper and its Streaming Variants

1. Core Architecture / Standard Whisper Encoder-Decoder

Robust Speech Recognition via Large-Scale Weak Supervision

4

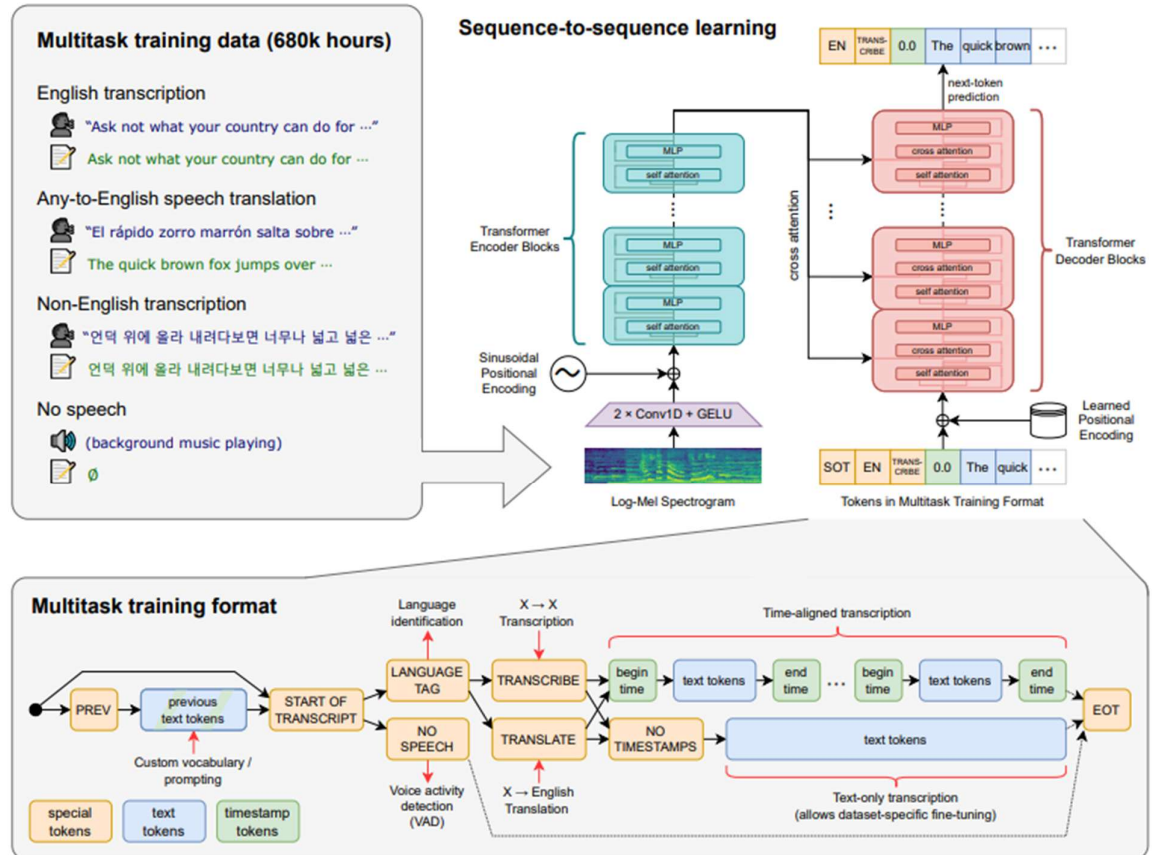


Figure 1. Overview of our approach. A sequence-to-sequence Transformer model is trained on many different speech processing tasks, including multilingual speech recognition, speech translation, spoken language identification, and voice activity detection. All of these tasks are jointly represented as a sequence of tokens to be predicted by the decoder, allowing for a single model to replace many different stages of a traditional speech processing pipeline. The multitask training format uses a set of special tokens that serve as task specifiers or classification targets, as further explained in Section 2.3.

Source: *Robust Speech Recognition via Large-Scale Weak Supervision (2022)*

2. ASR Block Processing – Sliding Window and local agreement

- How continuous microchunks of live audio concat into 30 sec sliding window to satisfy whisper's encoder requirement without massive latency.
- This involves local agreement policy where we compare tokens generated across overlapping segments.

Source: *Efficient Whisper on Streaming Speech (2023)*

3. Resource Efficient Streaming Pipelines (VAD + Buffer)

- may place a fast vad before whisper to filter out clinical noise, background hospital environment or irregular breathing patterns.

- Maps out energy based filter, dynamic buffer mechanism and adaptive chunk processing to prevent loss of info at speech boundaries.

Source: *WhisperPipe: A Resource-Efficient Streaming Architecture for Real-Time Automatic Speech Recognition*

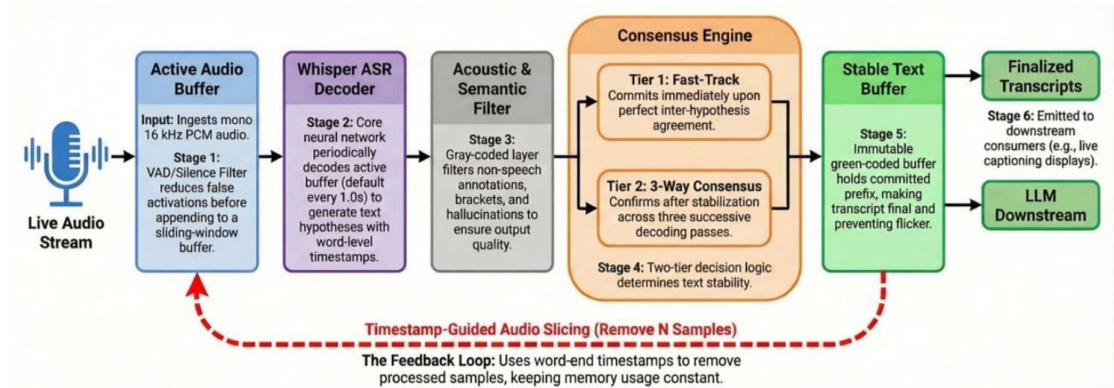


Figure 1. Overview of the WhisperPipe streaming pipeline. Audio is buffered, decoded by Whisper, filtered, and finalized using a two-tier consensus mechanism with timestamp-guided buffer management for efficient real-time transcription.

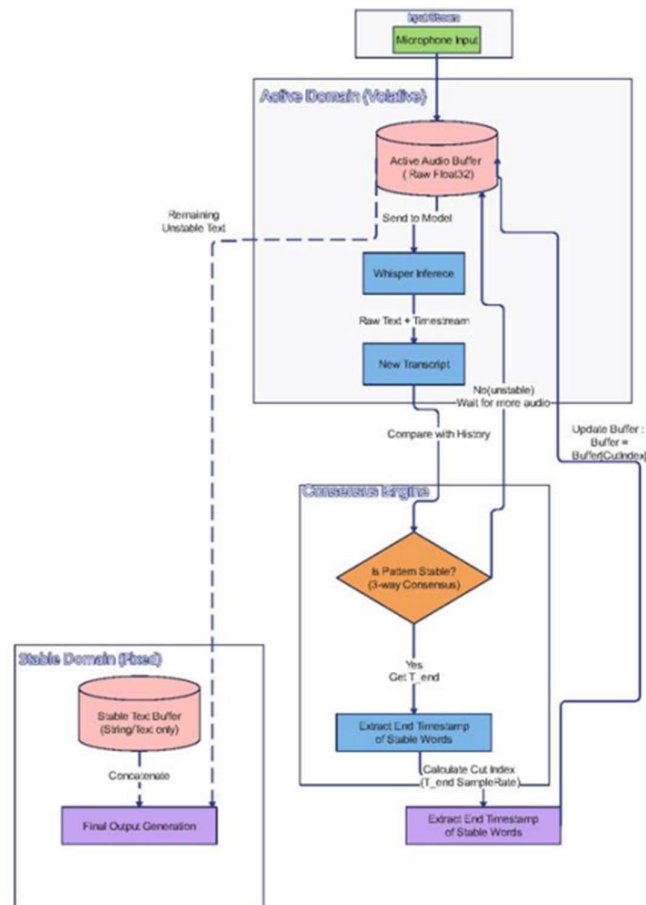


Figure 2. WhisperPipe's dual-buffer mechanism appends audio, commits stable text via a consensus engine, and trims the active buffer at the last committed timestamp to keep the decoding window bounded.

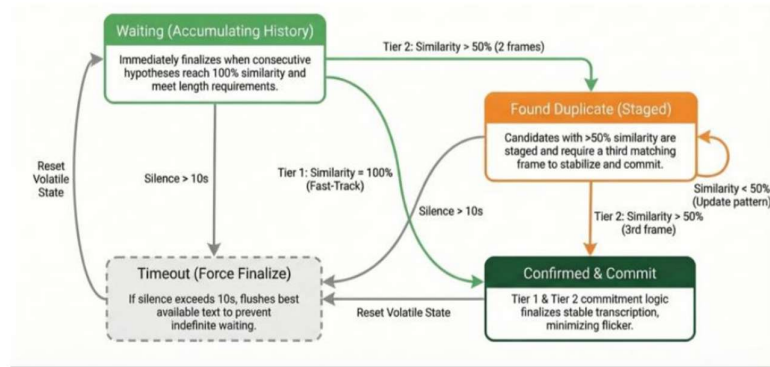
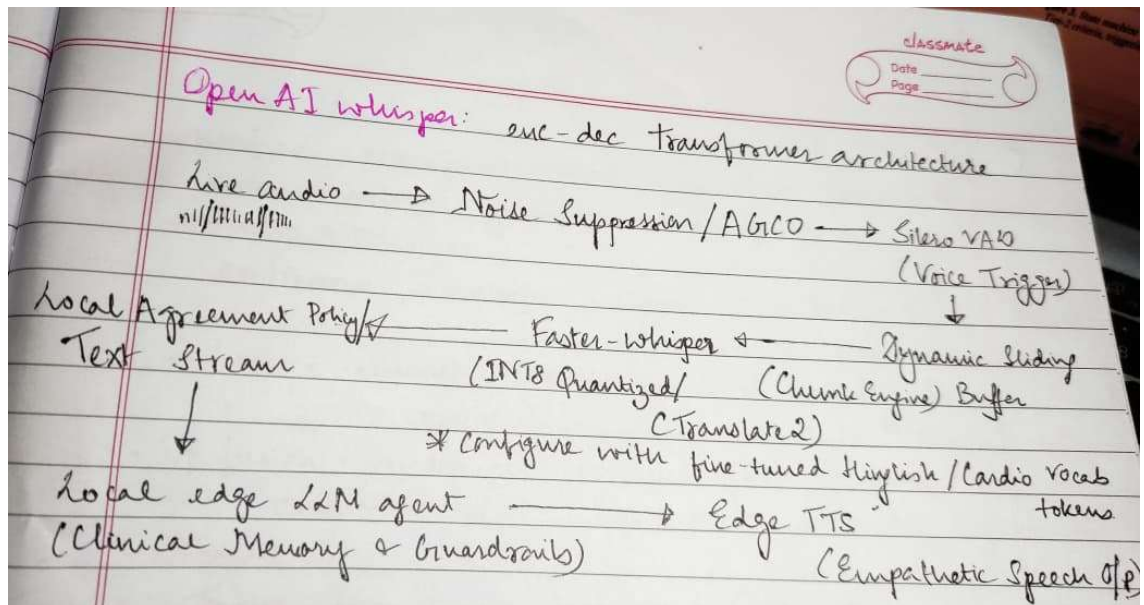


Figure 3. State machine of WhisperPipe's two-tier commit policy, where hypotheses accumulate until agreement satisfies Tier-2 criteria, triggering a 3-way confirmation to commit stable text. A timeout fallback finalizes the best available hypothesis to prevent indefinite waiting.

4. A possible chalked out pipeline using Whisper-



Proposed Pipeline-

